Conservative Routing Under Synthetic Shift: A Controlled Olympus Successor Study

A Preprint

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21 October 2026

Abstract

We evaluate a bounded successor to five unsuccessful Olympus routing primary endpoints. The model blends a frozen, development-selected reference with a learned alternative using an example-dependent intervention penalty and worst-group training. Seven correction variants and strong fixed, uniform, single-expert, direct, and constant-blend controls isolate competing explanations. Across five generated datasets and two optimizer seeds per dataset, adaptive robust routing achieves 99.619% accuracy on a reserved synthetic shift, versus 99.697% for the development-selected control. The paired difference is -0.078 percentage points (descriptive 95% dataset-cluster interval $[-0.410, 0.117]$), failing the prospectively required two-point gain. All familiar-regime nonregression criteria pass. The method improves familiar corruption accuracy by 0.605 points over its original reference, but gains no advantage over stronger continued-training controls. Under reserved shift it rescues nine reference errors while introducing 22 new errors across ten paired runs. The control's near-ceiling accuracy leaves only 0.303 points of possible gain, limiting the benchmark's ability to demonstrate the registered practical improvement. All forty protected reports, checkpoints, raw predictions, and exact replay checks are retained. The outcome is negative for the registered claim, not a general impossibility result or a full-architecture capability test.

Keywords mixture of experts · stacking · synthetic distribution shift · negative results

1 Introduction

Learned routing can degrade an already strong ensemble when the gate learns a relation between confidence, representation, and correctness that does not survive a change in observations. In the preserved Olympus sequence, five routing primary endpoints failed their declared improvement criteria. Some secondary interventions helped, including expanded corruption coverage and a residual temporal predictor. These observations motivate a narrower question: can a learned correction retain a strong reference while making useful, example-dependent changes under a reserved synthetic perturbation?

Our testable statement is: existing Olympus gates struggle to outperform strong controls under changed corruption configurations; a penalized, example-dependent blend around a frozen reference may improve the rescue-to-harm balance because it can reduce intervention where the alternative is unreliable. This is a hypothesis, not a guarantee. We compare anchoring, input dependence, and worst-group training in a fixed study with fresh source identities. The contribution is an empirical decomposition and reproducible negative-result continuity, not a new universal mixture architecture.

2 Related work and novelty boundary

Input-dependent expert mixtures are established by Jacobs et al. [1]. Learning a combiner from held-out predictions is the premise of stacked generalization [6]. Our two frozen banks and disjoint meta-fitting partitions implement that principle conservatively; they are not nested cross-fitting. The proposed convex blend is also ordinary probability mixing.

Sagawa et al. [5] show that small worst-group training loss does not ensure worst-group generalization and emphasize regularization. Our maximum of three minibatch losses is distinct from their adaptive-weight group-DRO algorithm. Ovadia et al. [4] motivate checking calibration under shift rather than trusting confidence alone. Neither paper proves improvement for this generator or this gate.

Learning to defer [3] studies a classifier's decision to use an expert with a consistent surrogate. We instead train a soft probability mixture and do not inherit its consistency theorem. Recent multi-expert deferral work identifies expert-selection ambiguity and underfitting [2]; it is a related concern, not an established explanation of our results. A direct comparison would require a separately specified deferral setting. We do not claim superiority over those algorithms or state of the art.

3 Problem formulation and scope

Each observation comprises eight integer tokens, a $3 \times 8 \times 8$ image, four five-dimensional observed states, and four three-dimensional actions. The target $y \in \{0, 1, 2, 3\}$ is a synthetic latent class. Domain metadata is sampled independently of the target; all four transitions are allowed and the supplied risk score is zero. No corruption identity or target is supplied to the gate. Three development groups describe clean, familiar-corruption, and known-shift observations. Group labels enter the objective only.

Five components produce class probabilities $p_j \in \Delta^3$ and states $z_j \in \mathbb{R}^{16}$. For two separately initialized banks trained on parity halves of a bank-fitting partition, we average corresponding probabilities and states. Thus $p_j = (p_j^{(0)} + p_j^{(1)})/2$ and similarly for z_j. The router receives

$$x = \mathrm{concat}(z_1, \ldots, z_5, p_1, \ldots, p_5, H(p_1), \ldots, H(p_5)) \in \mathbb{R}^{105}, \qquad H(p) = -\sum_c p_c \log \max(p_c, 10^{-8}). \tag{1}$$

These features are detached. Identical bank averaging is used during fitting, selection, and protected inference. No additional labels, groups, or corruption metadata are available at inference.

The study concerns four-class classification under this generator. It does not test language quality, tool use, autonomous research, natural-image reasoning, deployment safety, or six-family model promotion. The full system contains auxiliary heads that this objective does not supervise; their existence is not evidence that they acquired useful capabilities.

4 Model and chronological training specification

4.1 Inherited component banks

Each bank uses the repository's unchanged reference system. Text branches have separate weights but share the following architecture. Each text backbone has vocabulary 32, width 16, one pre-normalized causal attention block, four attention heads, a GELU feedforward layer of width 32, maximum length eight, and zero dropout. Token and positional embeddings are added; attention and feedforward sublayers have residual connections and a final layer normalization. Hermes uses this text representation. Prometheus adds domain-specific residual adapters of bottleneck width eight across three domains, with supplied domain IDs determining adapter selection. Perseus has four width-32 SiLU experts and top-two routing. Atlas combines text with 2×2 image patches through four-head cross-attention and maps its pooled representation to a normalized 12-dimensional retrieval vector. Kronos uses a width-12 GRU on concatenated five-dimensional states and three-dimensional actions. Its linear next-state head predicts all four next states. Aion is present in the inherited system but is not one of the five class experts.

Last-token Hermes, Prometheus, and Perseus states, the Atlas retrieval vector, and the last Kronos hidden state each pass through a linear projection and LayerNorm to width 16. Five separate $16 \to 4$ heads produce

logits. For each bank, 250 updates optimize

$$L_{\text{bank}} = \frac{1}{5B}\sum_{i,j} \text{CE}(\ell_{ij}, y_i) + 0.25 \text{ mean}_{i,t,k}(\hat{s}_{i,t+1,k} - s^*_{i,t+1,k})^2. \tag{2}$$

There is no counterfactual-consistency penalty. Each update samples 64 examples with replacement from that bank's 512-example parity half. The two banks are then frozen. The exact inherited operators and configuration are distributed with the source; no external model weights are used.

4.2 Reference fitting and selection

The semantic gate is $w(x) = \text{softmax}(W_2 \,\text{GELU}(W_1 \,\text{LN}(x) + b_1) + b_2)$, with widths $105 \to 32 \to 5$. Its prediction is $p_{\text{sem}} = \sum_j w_j p_j$. A fixed gate learns five logits, initialized at zero. On a separate reference-fitting partition, both receive 300 updates minimizing the sum of their mixture NLLs. Neither loss trains the frozen banks. Uniform averaging and each single expert are additional choices. An independent reference-selection partition chooses the highest minimum group accuracy, then the highest mean group accuracy, then earliest declared candidate order. The eight candidates are uniform, fixed, semantic, and experts 0 through 4. The resulting p_0 is frozen before correction fitting.

4.3 Alternative, intervention, and loss

Each alternative starts from a copy of the semantic reference network with gradients explicitly enabled. Let q_θ denote its mixture. The final prediction is

$$p_\theta(x) = (1 - \alpha_\theta(x))p_0(x) + \alpha_\theta(x)q_\theta(x). \tag{3}$$

Direct controls fix $\alpha = 1$. Constant controls learn a scalar logit. Adaptive controls use $\alpha = \sigma(a^T x + b)$ with $a = 0$ and $b = \log(0.1/0.9)$ initially. Constant controls also initialize at 0.1. The alternative network has 3,767 parameters, with zero, one, or 106 additional intervention parameters. This is a transparent capacity difference, not a perfectly parameter-matched comparison.

For group g, L_g is mean $-\log \max(p_{\theta,y}, 10^{-8})$ over 32 sampled examples. The ERM objective averages L_1, L_2, L_3; the robust objective takes their maximum. Constant and adaptive penalized variants add $0.01 \,\text{mean}(\alpha)$. The seven variants are direct ERM, direct robust, adaptive ERM, adaptive robust, constant ERM, constant robust, and adaptive robust without penalty. All receive 300 updates using the same per-group sample sequence on the correction-fitting partition. Adaptive robust is the prospectively fixed primary candidate.

All optimization uses AdamW, learning rate 0.001, weight decay 0.01, default Adam moments $(0.9, 0.999)$ and $\epsilon = 10^{-8}$, and global gradient norm clipping at one. Nonfinite losses, admitted gradients, or parameters terminate the run. Shared random seeds fix initialization and sampling. Inference is argmax over four class probabilities, with the framework's first-index tie rule. Probabilities must be finite, nonnegative, and sum to one within absolute tolerance 10^{-5}; intervention must lie in $[0, 1]$.

4.4 What anchoring does and does not prove

For any fixed example, equation (3) gives

$$\text{TV}(p_\theta, p_0) = \alpha \,\text{TV}(q_\theta, p_0) \le \alpha. \tag{4}$$

This follows by factoring the nonnegative scalar out of the ℓ_1 norm. It bounds distribution movement, not error: a small change can flip a narrow classification margin. A penalized mean intervention is not a hard per-example constraint. If R counts wrong-reference/correct-candidate cases and H counts correct-reference/wrong-candidate cases, then exactly

$$\text{Acc}(p_\theta) - \text{Acc}(p_0) = (R - H)/N. \tag{5}$$

Useful intervention requires $R > H$. There is no theorem that training NLL, worst-group loss, or the penalty produces that inequality under shift.

4.5 Inference cost

All arms use the same two banks. A routing network costs $O(105 \cdot 32 + 32 \cdot 5 + 5 \cdot 4)$ operations per example, plus normalization and activations, per routing network. An anchored arm with a semantic reference

evaluates both reference and alternative routing networks. Adaptive intervention adds one 105-dimensional dot product. The banks are expected to dominate, but no component-level timing was measured. The experiment retains all variants for comparison; a deployed single arm would not require all seven. We report observed runtime without claiming a general scaling law or measured FLOPs.

5 Data generation, splits, and perturbations

Labels are uniform across four classes. Initial clean state is independent Gaussian noise of standard deviation 0.15 with 1.5 added at the label coordinate. Actions are uniform on $[-1, 1]^3$. Clean dynamics are

$$s^*_{t+1} = 0.82 s^*_t + 0.18 \tanh(\mathrm{roll}(s^*_t, 1)) + 0.12 \,\mathrm{pad}(a_t, 2). \tag{6}$$

Observed states add Gaussian noise with standard deviation 0.12. Tokens start uniformly in $\{8, \ldots, 30\}$, with the first three replaced by $y + 1$, $5 + (y \bmod 2)$, and $7 + \lfloor y/2 \rfloor$. Images have uniform background noise below 0.15, with 0.75 added to a label-dependent quadrant and channel. These deliberately redundant class signals make this a mechanism diagnostic.

Broad bank training uniformly samples four branches: clean, one missing modality, one damaged modality, or one missing plus a different damaged modality. Missing means zero. Familiar damage randomizes the first three tokens, replaces the image by uniform noise, or replaces states by Gaussian noise. For half the broad examples selected independently, damaged observations instead use reversed first-three tokens, image inversion, or a 1.5 state bias. The clean development group has no imposed missingness. Corrupted and known-shift groups each have one missing and one distinct damaged modality; known shift uses reversal, inversion, or bias.

The protected composite-shift configuration uses the same missing/damaged assignment, but damaged text is rolled by two tokens, damaged images have alternating rows replaced by 0.5, and damaged states become $0.5s + \mathrm{linspace}(-0.75, 0.75, 4)$. Only the designated damaged modality gets its transform. These perturbation types were explored historically; their reserved configuration is not a previously unknown mechanism or natural data OOD. Targets and source IDs are unchanged by observation transformations.

Five dataset seeds are 20267100, 20267200, 20267300, 20267400, and 20267500. Each is repeated with training seeds 20267601 and 20267602. Bank initialization uses training seed plus $100b$ for bank b; bank sampling adds 10000. Reference sampling adds one. Correction construction resets the RNG to the training seed plus two, but alternative weights are copied from the trained semantic reference and intervention parameters are set to the stated constants. Correction sampling adds three. The second optimizer seed shares each dataset's examples and is not a sixth independent dataset.

Table 1: Disjoint stages per dataset. Three-regime stages use consecutive offsets.

Stage	Seed offset	Examples
Bank fit (split into two parity halves)	1	1,024
Reference fit	10–12	3×256
Reference selection	20–22	3×256
Correction fit	30–32	3×256
Final control selection	40–42	3×256
Protected four regimes	60–63	$4 \times 1,024$

Source ID equals generator seed times 100000 plus example index. All 85 dataset-stage seed ranges are disjoint; each stage records IDs and tensor hashes. The same protected tensors are used by all arms. Generated data have no human subjects or external dataset licenses; the repository's code license applies to its generator. Synthetic evidence cannot establish real-world rights, privacy, safety, or capability properties.

6 Frozen experimental methodology

6.1 Controls and selection

After correction fitting, the separate final-selection partition chooses among uniform, fixed, semantic, direct ERM, direct robust, constant robust, constant ERM, and experts 0–4, in that order. Selection maximizes worst-group accuracy, then mean accuracy, then declared order. This selected control is fixed before

protected evaluation. It is not the best control chosen after seeing test results. Every direct control receives the extra 300 updates, making additional training an explicit competing explanation. Adaptive ERM separates robust training from anchoring; constant controls separate input dependence from shrinkage; the no-penalty arm tests the regularizer.

6.2 Primary rule and uncertainty

The primary contrast is adaptive robust minus selected control accuracy on composite shift. Support requires mean gain at least 0.02 and descriptive 95% interval lower bound strictly above zero. For each clean, corrupted, and known-shift regime, nonregression against both selected control and reference requires mean delta at least -0.01 and interval lower bound at least -0.02. All seven criteria must hold. A primary failure is retained regardless of any favorable secondary metric or arm.

For each dataset, the two optimizer-seed paired deltas are averaged. The five dataset deltas are then resampled with replacement 10,000 times using NumPy's default generator seeded 20267999. The mean of five sampled deltas gives each bootstrap replicate; percentile 2.5 and 97.5 bounds are descriptive pilot intervals. Five clusters are too few for strong general population claims. All per-dataset deltas are reported. Secondary comparisons are exploratory, without multiplicity-adjusted significance declarations.

Metrics include accuracy, NLL, summed-class Brier score, ten-bin expected calibration error, top-half confidence selective accuracy, intervention amount, prediction changes, rescues, and harms. We retain individual expert predictions. The rescue identity is checked arithmetically. Primary and nonregression rules are fixed before execution; no protected result may select a new primary.

6.3 Execution, freeze, and failure policy

One campaign contains ten development cells, twenty bank fits, seven correction fits per cell, and forty protected reports. There are zero scientific development revisions, zero paid jobs, and no external data or weight downloads. The environment is a locked CPU installation with one Torch thread and deterministic algorithms. A 120-second artificial qualification must pass on the exact scientific source. The campaign's hard subprocess deadline is 3,600 seconds. Resource checks require at least 2 GiB available memory and at most 90% swap use; macOS additionally requires 512 MiB raw free memory and stable pageouts. Failures stop and retain partial artifacts.

Before generating protected data, the freeze binds the protocol, source hashes, Python/Torch/platform contract, all development file identities, reference and control selections, checkpoints, and split hashes. Qualification tests include convex endpoints, gradients after freezing a reference, hand-calculated group losses, disjointness, tie-breaking, complete tiny lifecycle, and deliberate tampering. Verification regenerates every protected split, reloads both banks and all routers, exactly replays probabilities and intervention, and recomputes metrics and summaries. During development, checkpoint reload separately regenerates final-selection features and compares predictions exactly. Other development partitions retain IDs and tensor hashes; verification does not replay their training trajectories. This is computational replay, not independent retraining.

6.4 Prospective limitations and falsifiers

A positive result would support only this bounded synthetic configuration. Correlated experts may leave little corrective opportunity; the selected reference may already be strong; the intervention penalty may suppress useful changes; and worst-group optimization can fit the wrong training surrogate. Our ablations distinguish some of these alternatives but cannot establish a complete causal explanation. No model/data-scale sweep, broad hyperparameter sensitivity study, or external benchmark is admitted. Such work requires a successor protocol rather than post-hoc expansion of this test. Invalid provenance or numerical behavior invalidates the test; absence of execution means not evaluated, not negative performance.

7 Execution and evidence integrity

The campaign ran once on a free GitHub-hosted Ubuntu CPU runner from commit `18a7052`, using Python 3.14.8 and Torch 2.14.1+cu130 in CPU mode. It completed ten development cells, twenty bank fits, and forty protected reports in 248.57 seconds, including replay. Development took 22.26–22.92 seconds per cell. Initial available memory was 15.52 GB, with zero swap use. The admission receipt records an initial

process peak RSS of 35.30 MB before Torch import; this is not campaign peak memory. Periodic resource checks passed, but a full peak-memory series and FLOP profile were not retained. We do not claim those measurements.

Kaggle was the requested platform. Its authenticated draft editor remained in an editor-loading state with execution unavailable. The same frozen CPU protocol ran on GitHub instead, a recorded platform deviation. No Kaggle v6 scientific result is claimed. A self-contained source-pinned Kaggle notebook defaults to artificial qualification; a second scientific campaign requires separate reproduction admission.

Fourteen artificial regressions passed on the exact scientific source before the campaign. Source and environment were bound before protected generation. The verifier regenerated all protected inputs and exactly matched retained bank, route, and intervention predictions, then recomputed all forty reports and the summary. The retrieved archive's SHA-256 matches the GitHub artifact digest. Separate tensor-free analysis checks file hashes, report means, five paired dataset deltas, and the rescue/harm identity, generating the tables and plot below. These are computational consistency checks, not independent training.

8 Protected results

8.1 Primary endpoint and nonregression

The registered result is NEGATIVE. Adaptive robust routing reaches 99.6191% composite-shift accuracy versus 99.6973% for the selected control. Its paired mean delta is -0.078125 percentage points, with descriptive 95% interval $[-0.410156, 0.117188]$. The five dataset deltas, after averaging two training seeds, are $[0, 0.097656, 0.146484, 0.097656, -0.732422]$ points. One dataset contributes the negative tail; the others offer at most a small positive difference. Both the required $+2$ points and positive lower bound fail.

All six nonregression comparisons pass. Against selected control, deltas are zero on clean, -0.009766 points on familiar corruption, and $+0.009766$ points on known shift. Against reference they are zero, $+0.605469$, and zero. Avoiding a large familiar-regime regression does not rescue the failed improvement claim. The intervals are pilot descriptions, not population safety certificates.

Table 2: Adaptive robust minus selected control, in percentage points. Intervals are paired five-dataset descriptive bootstrap intervals.

Regime	Mean	Lower	Upper
clean	0.000	0.000	0.000
corrupted	-0.010	-0.098	0.098
known shift	0.010	0.000	0.029
composite shift	-0.078	-0.410	0.117

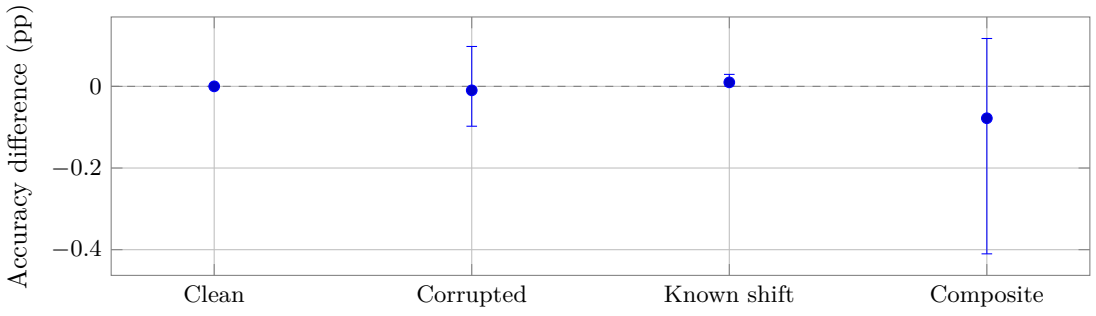


Figure 1: Adaptive robust minus the development-selected control. Error bars show descriptive 95% paired dataset-cluster bootstrap intervals. Only the composite contrast is the improvement primary; its practical threshold is $+2$ pp.

Table 3: Protected accuracy (percent), averaged across ten cells. Selection aliases vary by development cell; all arms share bank predictions.

Arm	Clean	Corrupted	Known shift	Composite
adaptive erm	100.00	93.56	99.99	99.67
adaptive robust	100.00	93.54	99.99	99.62
adaptive robust no penalty	100.00	93.57	99.98	99.58
constant erm	100.00	93.19	99.99	99.76
constant robust	100.00	93.19	99.99	99.77
direct erm	100.00	93.52	99.99	99.53
direct robust	100.00	93.51	99.98	99.54
expert 0	99.98	50.21	75.25	63.89
expert 1	99.99	49.71	75.21	62.64
expert 2	100.00	49.82	75.36	67.90
expert 3	100.00	69.87	84.07	94.69
expert 4	99.99	49.70	74.67	74.98
fixed	100.00	82.35	99.87	99.83
reference	100.00	92.94	99.99	99.75
selected control	100.00	93.55	99.98	99.70
semantic	100.00	92.94	99.99	99.75
uniform	100.00	78.28	99.50	99.38

8.2 Ablations and competing explanations

Reference selection chose semantic routing in all ten cells. Final control selection chose direct ERM five times, direct robust four times, and constant robust once. Improvement over the original reference can therefore arise from additional fitting on new correction data. The primary candidate does not establish an advantage over that competing explanation.

Adaptive ERM scores 99.6680% on composite shift, modestly above adaptive robust at 99.6191%. Constant robust reaches 99.7656%, and fixed routing reaches 99.8340%. These are descriptive secondary observations, not retrospectively selected primary winners. Fixed routing scores only 82.3535% on familiar corruption versus 93.5449% for adaptive robust, demonstrating regime dependence. Uniform routing scores 78.2813% on familiar corruption and 99.3848% on composite shift. Neither a global ranking nor the best protected arm replaces the frozen primary.

Removing the adaptive robust penalty yields 99.5801% composite accuracy, slightly below the penalized arm. Adaptive robust also exceeds direct robust there (99.5410%), while constant robust remains higher. These differences do not establish that example-dependent intervention or worst-group training is necessary. No adjusted secondary significance claim is made.

8.3 Probability quality and action quality differ

On composite shift, adaptive robust reduces reference NLL from 0.11085 to 0.10705 and Brier score from 0.02675 to 0.02562 while lowering accuracy. Its ECE is 0.09130 versus reference 0.09619. Selected control remains better on all three: NLL 0.10352, Brier 0.02448, ECE 0.08927. Probability quality can improve relative to a reference without producing more correct decisions or beating a stronger control. NLL improvement is not proof of an accuracy gain.

On familiar corruption, candidate NLL is 0.22352 versus reference 0.24456 and selected control 0.22013. Candidate Brier is 0.09981 versus 0.11134 and 0.09879; candidate ECE is 0.07376 versus 0.07699 and 0.06776. These secondary metrics reinforce the extra-training explanation rather than isolate adaptive superiority.

8.4 Intervention failure analysis

Mean mixture weight rises above its initial 0.1: approximately 0.441 on clean, 0.697 on familiar corruption, 0.609 on known shift, and 0.676 on composite shift. This is a probability-mixture weight, not a decision-change rate. The penalty does not keep mean mixture weight near its initial 0.1. Its coefficient is a soft tradeoff against NLL, not a hard constraint or correctness detector.

Table 4: Adaptive-robust intervention against the frozen reference. Counts sum ten cells (two initializations share each dataset).

Regime	Mean α	Rescues	Harms	Changes
clean	0.4410	0	0	0
corrupted	0.6969	130	68	215
known shift	0.6094	0	0	0
composite shift	0.6758	9	22	31

On composite shift, nine rescues and 22 harms yield $(9 - 22)/10240 = -0.00126953$ accuracy relative to reference, exactly the recorded -0.126953 percentage-point reference contrast. There are 31 changed decisions. On familiar corruption, 130 rescues and 68 harms yield $62/10240 = 0.00605469$, or $+0.605469$ points. Of 215 changed predictions, 17 change from one wrong answer to another. Counts include two initializations of the same five datasets and are not 10,240 independent experimental units. Clean and known-shift decisions are unchanged from reference despite substantial probability mixing.

Adaptive correction helps some familiar-corruption reference errors but introduces more errors than it repairs under reserved shift. The familiar gain is matched by simpler continued-training controls; no additional adaptive advantage is established. The evidence does not causally identify whether the penalty, representation, optimization, expert correlations, or shift configuration causes the failure. Those explanations require successor tests.

9 Benchmark ceiling and scientific interpretation

The selected control achieves 99.6973% on the protected primary regime. Even perfect prediction could improve that observed accuracy by only 0.3027 percentage points, below the registered two-point threshold. This ceiling was identified after protected inspection. We retain the threshold and negative result. Lowering the requirement or selecting a harder protected subset now would invalidate confirmation.

The study does not support the promised practical gain, and its point estimate is unfavorable. It also cannot establish the absence of a useful adaptive mechanism on harder, separately defined data. The test is valid as the registered comparison but has poor discriminatory scope for a two-point improvement in this near-saturated regime. A future benchmark should establish task difficulty and corrective opportunity on development evidence before freezing an untouched confirmation set.

10 Preserved predecessor chronology

Table 5: Preserved routing contrasts in percentage points. Endpoints differ; this is a chronology, not a common learning curve. All listed primaries failed.

Study	Endpoint/comparator	Delta	95% interval
v2	Corrupt selective / uniform	-4.69	$[-9.53, 0.16]$
v3	Corrupt selective / semantic	-1.25	$[-2.81, 0.31]$
v4	Corrupt accuracy / semantic	-6.82	$[-7.71, -5.89]$
v5	Reserved shift / core semantic	-0.068	$[-0.244, 0.107]$
EditCLEVR	Shifted / selected control	-0.005	$[-0.015, 0]$
v6	Composite / selected control	-0.078	$[-0.410, 0.117]$

Earlier ledgers and the consolidated manuscript remain unchanged. In v4, expanded corruption coverage improved semantic corrupted accuracy by 27.42 points, and semantic routing exceeded learned fixed routing by 10.21 points. Removing its counterfactual penalty changed the relevant comparison by only about 0.01 points. These are bounded secondary observations.

The separate v5 temporal primary remains supported within its synthetic task: clean residual MSE was 0.014566 versus 0.071332 for the preceding predictor, a 79.58% reduction. Residual-minus-persistence was -0.015398 with interval $[-0.015830, -0.014920]$. This is temporal-prediction evidence, not routing improvement. V5 and EditCLEVR raw remote archives remain pending local retrieval. Their recorded aggregate/provenance findings are preserved, not independently reconstructed in this study.

11 Discussion and next falsifiable direction

The positive engineering outcome is a functioning, source-bound experiment with strong controls and retained negative evidence. The narrower scientific observation is that a useful familiar-regime reference correction need not be a better shifted-regime policy. Direct mixture-NLL training combined with this conservative correction did not establish the desired advantage. This study does not isolate the earlier correctness-score-to-mixture proxy mismatch as the cause.

A defensible successor would first measure, using development data alone, which experts rescue each other's errors and whether a gate has information to predict those rescues. It would use a task whose baseline leaves room for the declared practical effect, compare to continued-training and constant controls again, and test a materially different mechanism for distinguishing rescue from harm. Harder inputs alone are not an improvement. Each change needs a predicted consequence and an untouched confirmation set. No such successor is executed or promoted here.

12 Limitations, ethics, and reproducibility

The five independent units are generator seeds, not independent natural-world datasets. Signals are synthetic and redundant; three text experts share an input channel. Perturbation types are historically known. Capacity differs between arms, intervals use only five clusters, and no model-size scaling, broad sensitivity sweep, real-world benchmark, or independent retraining is included. Saturation especially limits the practical-improvement test. These limits preclude general capability, safety, production, or state-of-the-art claims. Zero of six Olympus families is promoted.

Scientific source, predictions, checkpoints, JSON reports, protocols, and the verified archive accompany the artifact. The execution environment is recovered by checking out `18a7052` and its lockfile. Later publication commits patch Vite, certifi, and Pillow after release audits; they do not change frozen model, generator, protocol, or tests. Results are not silently attributed to the patched environment. The standalone renderer regenerates tables and PGFPlots source from retained JSON without Torch.

The repository is publicly viewable under its existing proprietary license; public availability does not grant unrestricted reuse. External reproduction or redistribution must respect that license or obtain the copyright holder's permission. No personal data or human-subject experiments are introduced. The supplied style is retained byte-for-byte and the predecessor's author display name is preserved. This manuscript is not submitted to arXiv or peer reviewed. Independent source and arithmetic review is distinct from independent experimental reproduction.

13 Conclusion

Conservative adaptive routing did not improve the frozen composite-shift primary over a strong development-selected control. Nonregression passed and familiar-corruption reference accuracy improved, but continued-training controls matched that gain. The registered study closes with a negative result, retained evidence, and a narrower successor question about corrective opportunity and rescue-versus-harm prediction.

A Artifact map and reproduction commands

The study directory is `research/olympus_routing_v6`. `protocol.json` is the authoritative contract; `campaign-evidence.zip` retains all raw tensors and checkpoints; `EVIDENCE_RECEIPT.json` records the externally verified archive digest. The unpacked `retained-campaign/routing-v6-campaign` contains the contract, scientific freeze, summary, verification receipt, and development and protected reports. Restore raw tensors from the ZIP after a Git clone.

For an authorized reproduction in the frozen checkout, use the locked environment and new output directories:

```
uv sync --locked --all-extras
.venv/bin/python -m scripts.run_conservative_routing_v6 qualify \
  --output artifacts/qualification-NEW
.venv/bin/python -m scripts.run_conservative_routing_v6 run \
  --output artifacts/campaign-NEW \
  --qualification artifacts/qualification-NEW
.venv/bin/python -m scripts.run_conservative_routing_v6 verify \
  --output artifacts/campaign-NEW
```

The public commands enforce resource and hard wall-time limits. Directly calling the internal worker is not an admitted reproduction procedure. An additional scientific execution must be separately recorded, not a search for a favorable replacement result.

B Review and correction trail

Pre-execution independent review found and repaired inherited frozen gradients in copied alternatives, incomplete bank checkpoint replay, acceptance of self-consistently rewritten predictions, and an unbound environment contract. Artificial tamper tests cover the repairs. Protocol versions 6.0.0 and 6.0.1 remain retained; 6.0.2 was frozen before scientific data generation. Prospective methods and their hashes are retained at the execution commit. Later wording corrections clarify copied weights, separate backbone parameters, protected-only replay, and inference-cost scope without changing the tested mechanism. The flaw audit records outstanding scientific limits alongside engineering repairs.

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